



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

# NPM PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Build & Package Management

TOTAL: 10 QUESTIONS

**Q1** What is npm and what UI problem does it address? [Threat Model](#)

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**Q6** How do you fetch data and handle loading/error states in npm? [Form](#)

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**Q2** How do components communicate in npm? [Architecture](#)

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**Q7** What performance techniques apply to npm applications? [Recycle](#)

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**Q3** How does npm manage state for complex applications? [Configuration](#)

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**Q8** How do you handle forms and validation in npm? [Compliance](#)

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**Q4** What rendering strategies does npm support? [Hardening](#)

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**Q9** How do you test npm components? [Testing](#)

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**Q5** How does npm handle client-side routing? [SSO / IdP](#)

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**Q10** What accessibility (a11y) practices are essential in npm? [Tuning](#)

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 npm is a component-based library or framework

Mitigation

npm is a component-based library or framework for building interactive user interfaces. It provides a reactive programming model where UI updates automatically when underlying state changes, reducing imperative DOM manipulation.

A6 Use a data fetching hook or library

Observability

Use a data fetching hook or library that returns { data, isLoading, error }. Show a skeleton or spinner while loading, an error message on failure, and the data on success. Avoid fetching in render to prevent waterfalls.

A2 Parent-to-child communication uses props (one-way data)

Initiatives

Parent-to-child communication uses props (one-way data flow). Child-to-parent uses events or callbacks. Siblings share state via a common parent or a global store (Redux, Pinia, Signals). Avoid prop-drilling by using context or state management.

A7 Code splitting loads only the code needed

Automation

Code splitting loads only the code needed for the current route. Memoization (useMemo, React.memo) prevents unnecessary re-renders. Virtualization renders only visible list items. Images use lazy loading and next-gen formats (WebP, AVIF).

A3 Local component state handles UI-only concerns. Shared

Rendering

Local component state handles UI-only concerns. Shared state (user auth, cart) lives in a global store with a predictable update pattern (actions, reducers, or mutations). Server state (API data) is managed by libraries like React Query or SWR.

A8 npm manages form state with controlled inputs

Governance

npm manages form state with controlled inputs or a form library (React Hook Form, VeeValidate). Validation runs on blur or submit, displaying field-level error messages. Schema libraries (Zod, Yup) define validation rules declaratively.

A4 npm supports CSR (browser renders), SSR (server

Gotchas

npm supports CSR (browser renders), SSR (server renders HTML on each request), SSG (pages generated at build time), and ISR (pages regenerated on a schedule or on-demand). Choose based on SEO requirements and data freshness needs.

A9 Unit tests check individual component rendering and

Assessment

Unit tests check individual component rendering and interactions using a component testing library. Integration tests verify multi-component flows. End-to-end tests (Playwright, Cypress) simulate real user journeys in a browser.

A5 npm's router (built-in or a library like

Federation

npm's router (built-in or a library like React Router) maps URL paths to components without page reloads. Dynamic segments (/users/:id) are parsed and passed as params. Guards or middleware can protect routes requiring authentication.

A10 Use semantic HTML (button, nav, main) instead

Optimization

Use semantic HTML (button, nav, main) instead of divs with click handlers. Ensure keyboard navigability (tab order, focus management). Add ARIA roles and labels for dynamic content. Test with a screen reader and run axe or Lighthouse audits.